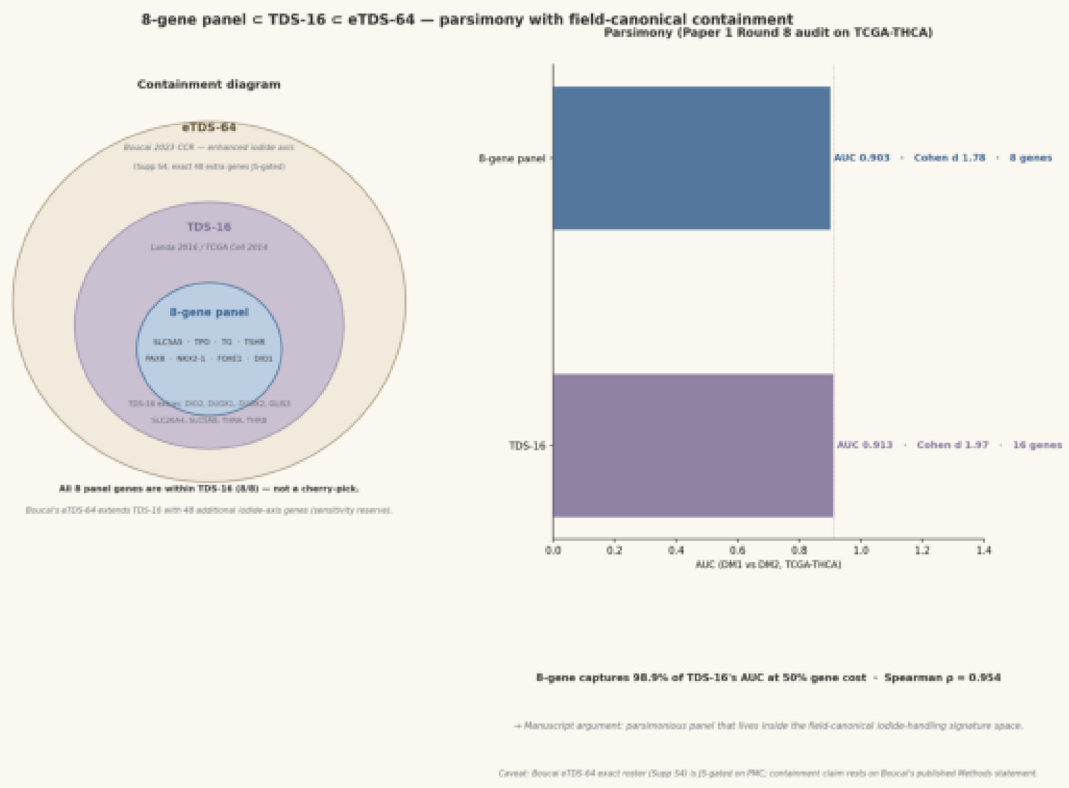
Reviewer-defense scorecard for the 8-gene RAI differentiation panel

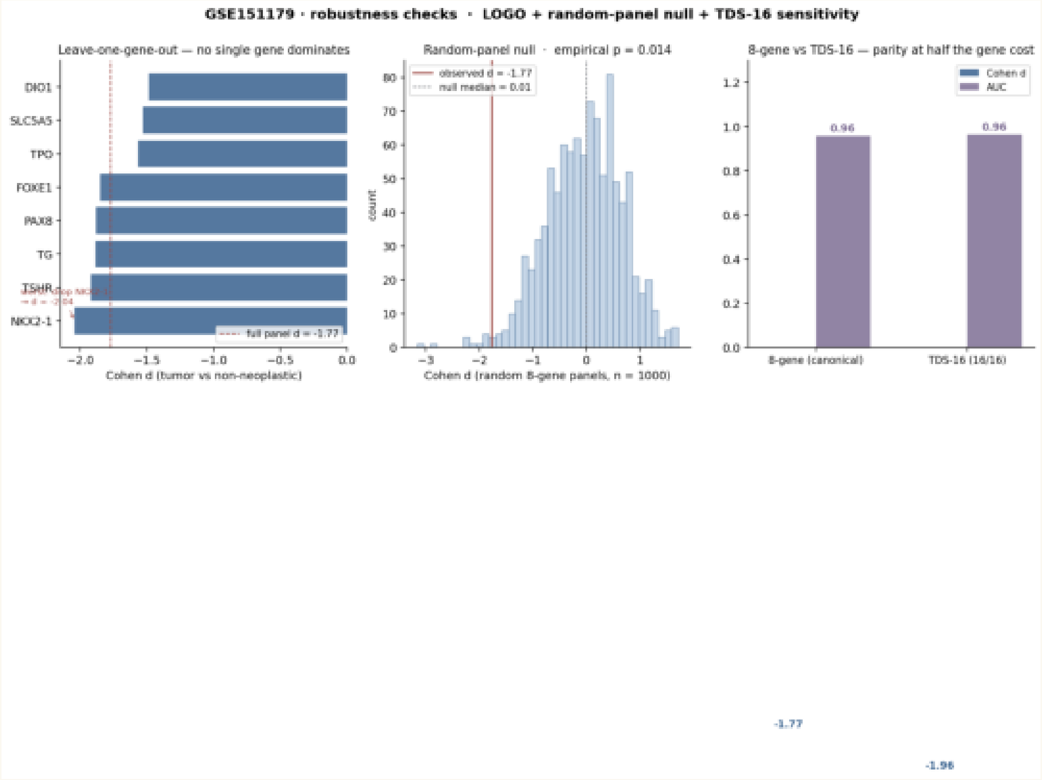
Four reviewer-ask attack vectors with co-located evidence · one image

a · Panel ⊂ TDS-16 ⊂ eTDS-64 containment + parsimony



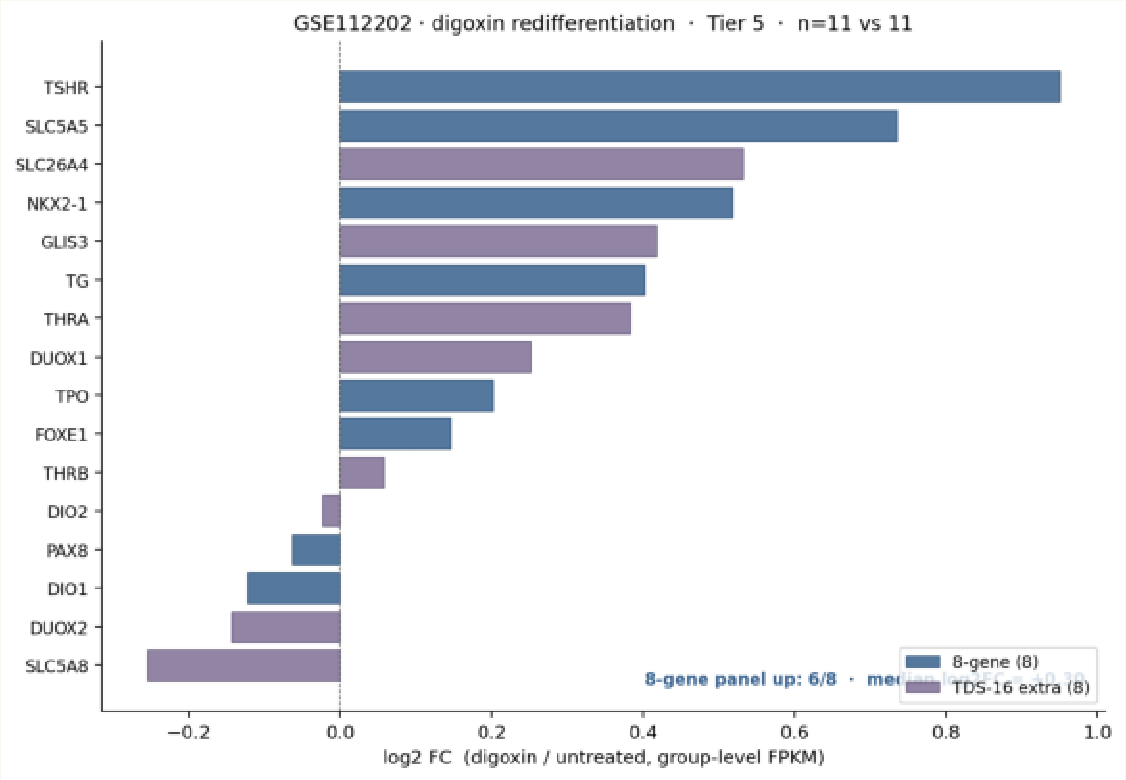
8/8 panel genes are within TDS-16; 8-gene captures 98.9 % of TDS-16 AUC at 50 % gene cost.
Spearman ρ = 0.954 (Paper 1 R8). Reviewer R2 cherry-pick defense layer 1.

b · GSE151179 robustness — LOGO + random-panel null + TDS-16 sensitivity



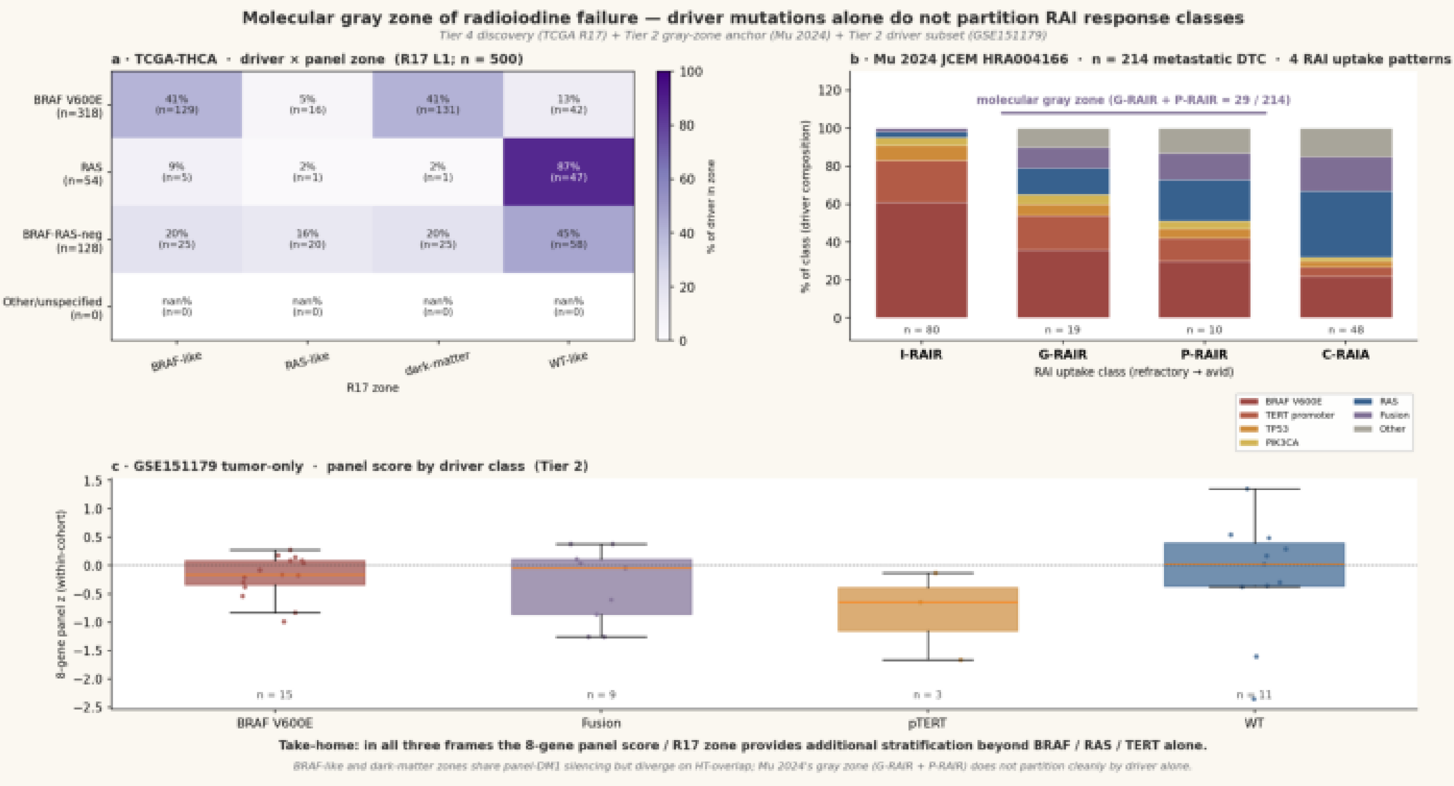
LOGO d ∈ [−2.04, −1.49]; 1000-permutation empirical p = 0.014;
TDS-16 d = −1.96, AUC = 0.96. Reviewer R2 defense layer 2-4.

c · GSE112202 redifferentiation (Tier 5) — direction-of-effect



6 / 8 panel genes upregulated by digoxin; median log2FC = +0.30.
TSHR +0.95, SLC5A5 +0.73, NKX2-1 +0.52, TG +0.40. Mechanism plausibility.

d · Driver × zone × Mu 4-class gray zone



Mu 2024 gray zone (G-RAIR + P-RAIR) = 29 / 214 = 14 %; drivers do not partition cleanly.
BRAF-like and dark-matter zones share panel silencing but diverge on HT-overlap. R3 defense.